

AUTONOMOUS RC CAR Final PRESENTATION

Brendan Schuler, Nicholas Paz, Stan Daaboul

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.



Applications

Instead of sending people into dangerous unstructured environment, an autonomous vehicle can be much safer and resourceful to use with ai detection. Some unstructured environments that it can be applied to:

- Dense forests: for forestry pursuits
- Caves and Mines: can be used for mining ventures
- Damaged nuclear power plants: to assess rubble or even clean up the site
- Natural disasters: finding survivors through rubble after earthquakes or destructive storms



Initial steps/goals we had for our project

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

Step 03

Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

Step 04 (Bonus)

Can it create paths based on given targets?

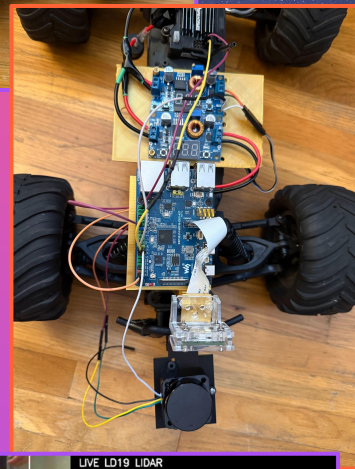
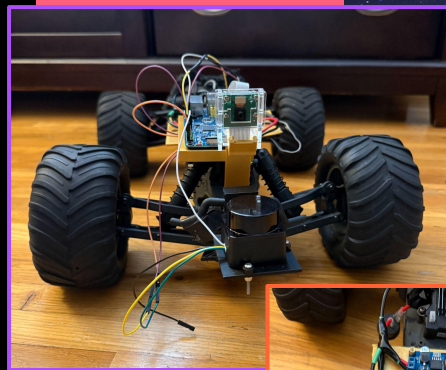
Now that our project is done, what is the highest step we were able to achieve?

Current Project Deliverables

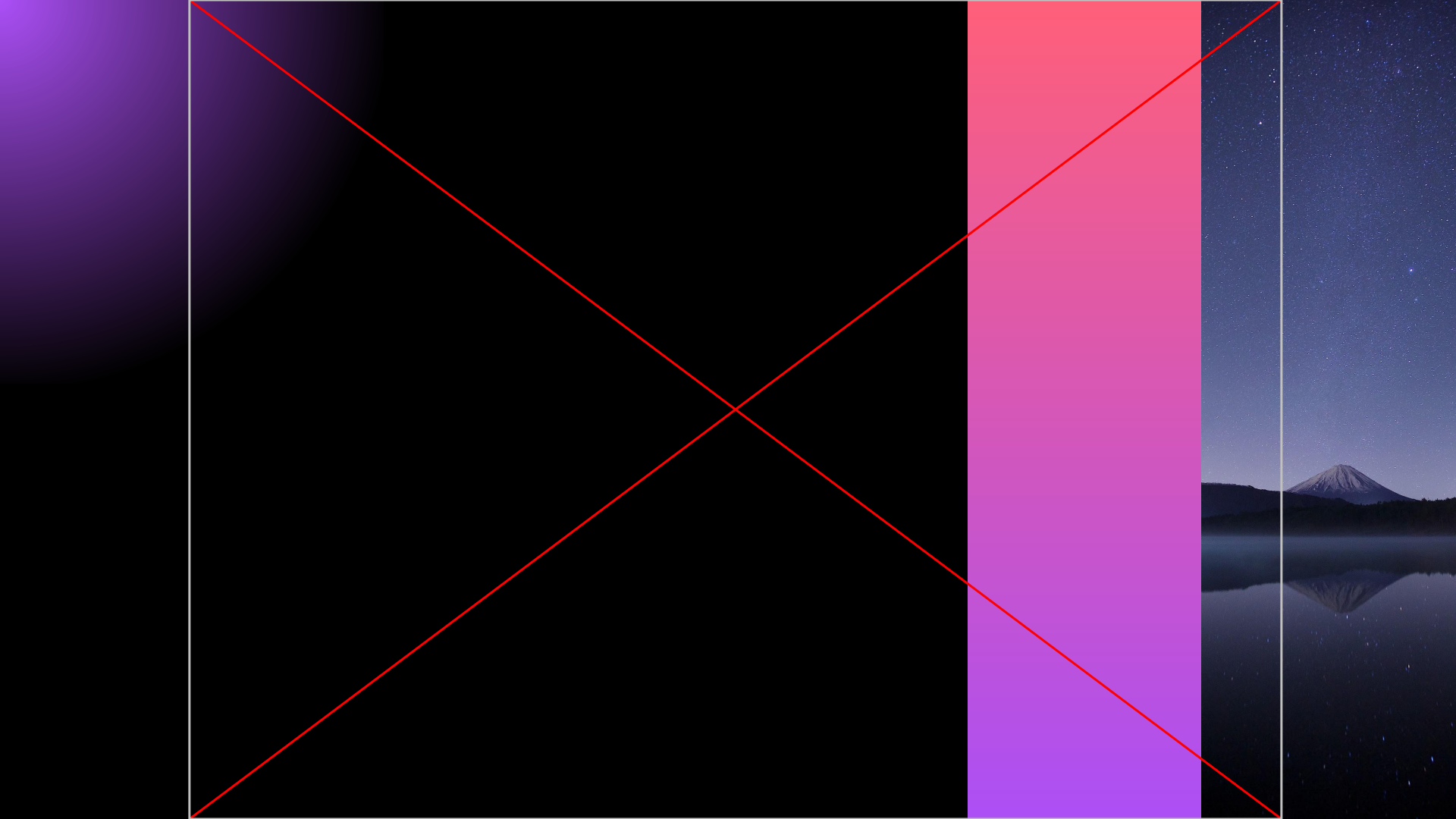
- Working modified car that can avoid obstacles successfully
- An AI software that is successful in detecting objects like trees, people, traffic cones, and bushes

Some Issues That are Still Unsolved

- Frame is too small for serious off-roading ventures, would need a larger base that can handle seriously dicey terrain.
- Some parts are of low quality and could be replaced for smoother obstacle avoidance
- Would need parts to be more securely assembled onto the frame before it is consumer ready.

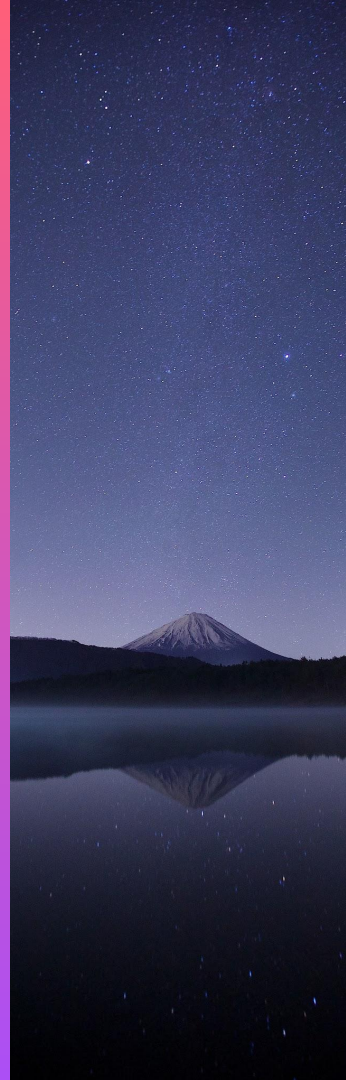


Demo of Obstacle Avoidance



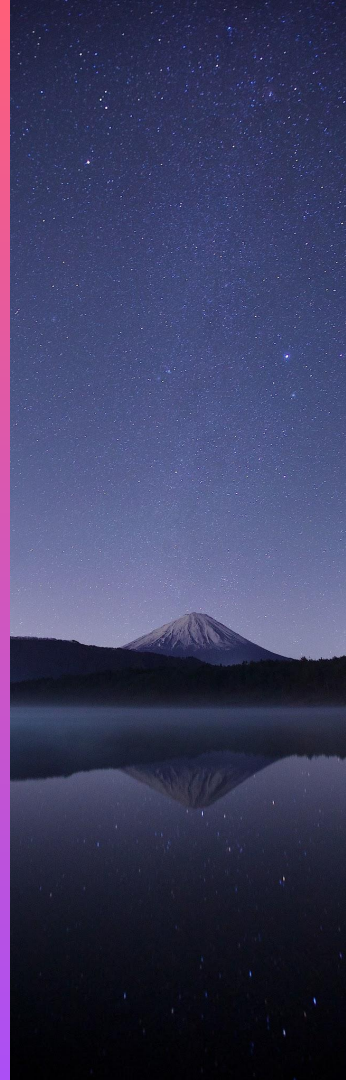
Future Prospects

- The next steps are for the AI to be trained for more obstacles, for the physical design to be optimized, and to make not just the turning but also the throttling autonomous.
- The basic framework we have built that connects the electronics, software, and physical components gives future students the opportunity to improve on the software right away and focus more on the algorithm aspect, which will bring us closer to the initial vision of this project.



Lessons Learned

- In a project like this, you don't develop only one aspect at a time, like the software OR the hardware OR the electronics. It is a constant process of finding the current biggest limiting factor to the robot's functionality, and working on it.
- You don't need the most expensive and high quality parts you eventually want to use in order to make progress on a project. Finding an MVP you can test and gather data from before investing fully is valuable.
- Accomplishing something tangible every day is the only way to make progress. It's easy to get caught up in trying to plan everything from the beginning, but ultimately you have to be prepared to just identify what is the next immediate step on any given day, regardless of how the plan is going.



THANK YOU

Any questions?